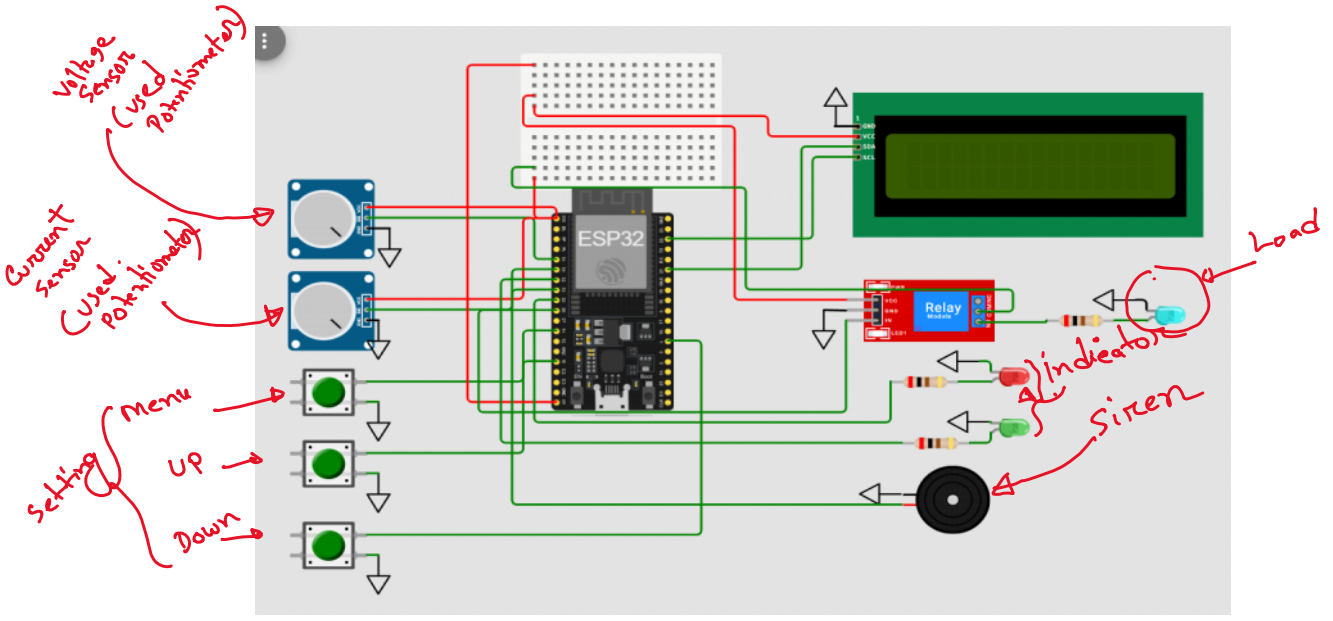
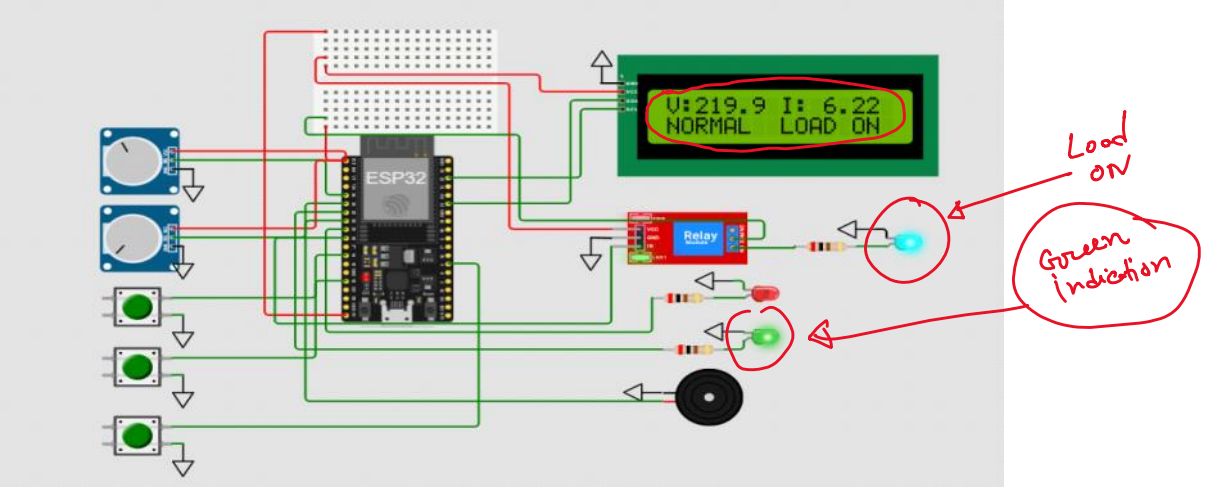
Home_Appliances_Protector

Saturday, October 10, 2026 3:42 PM

Simulation Circuit Diagram:



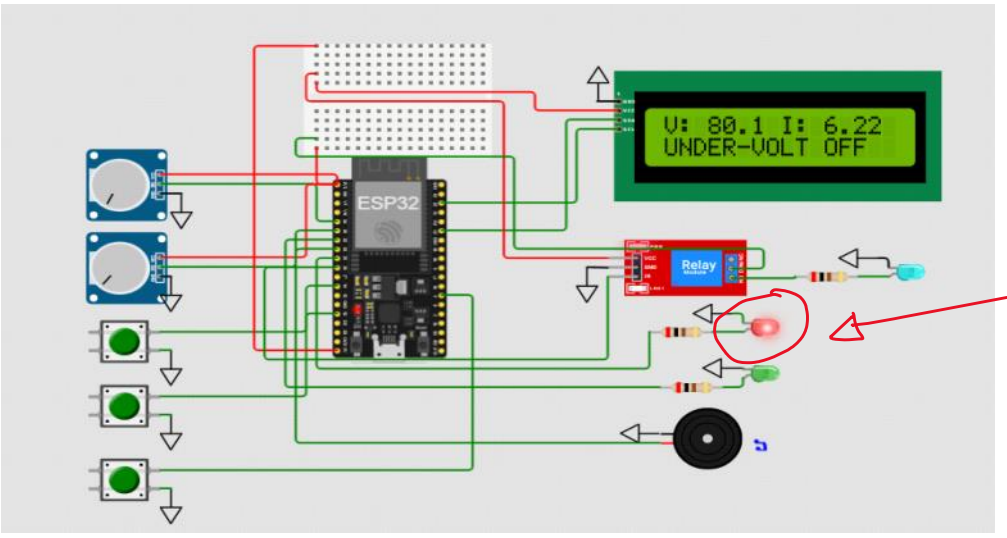
Normal Voltage & Normal Current Condition:



Result in Serial Monitor:

```
[13001 ms] V=219.9 V  I=6.22 A  LOAD=ON  STATE=NORMAL  FAULT=NONE  TRIPS=0
[14017 ms] V=219.9 V  I=6.22 A  LOAD=ON  STATE=NORMAL  FAULT=NONE  TRIPS=0
[15033 ms] V=219.9 V  I=6.22 A  LOAD=ON  STATE=NORMAL  FAULT=NONE  TRIPS=0
[16049 ms] V=219.9 V  I=6.22 A  LOAD=ON  STATE=NORMAL  FAULT=NONE  TRIPS=0
[17065 ms] V=219.9 V  I=6.22 A  LOAD=ON  STATE=NORMAL  FAULT=NONE  TRIPS=0
[18081 ms] V=219.9 V  I=6.22 A  LOAD=ON  STATE=NORMAL  FAULT=NONE  TRIPS=0
[19097 ms] V=219.9 V  I=6.22 A  LOAD=ON  STATE=NORMAL  FAULT=NONE  TRIPS=0
```

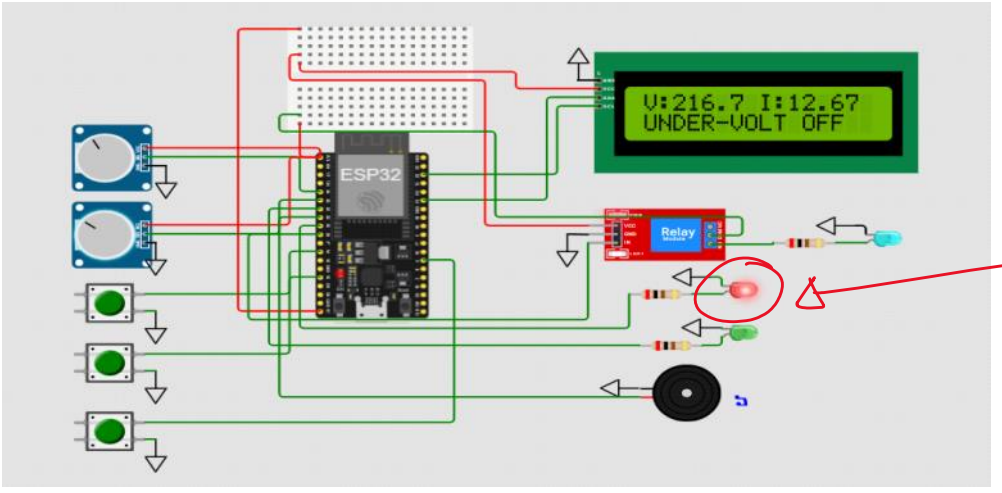
Low Voltage Condition:



Result In Serial Monitor:

```
[26206 ms] V=161.0 V  I=6.22 A  LOAD=OFF  STATE=DISCONNECTED  FAULT=UNDER-VOLTAGE  TRIPS=1
[27222 ms] V=80.1 V  I=6.22 A  LOAD=OFF  STATE=DISCONNECTED  FAULT=UNDER-VOLTAGE  TRIPS=1
[28238 ms] V=80.1 V  I=6.22 A  LOAD=OFF  STATE=DISCONNECTED  FAULT=UNDER-VOLTAGE  TRIPS=1
[29254 ms] V=80.1 V  I=6.22 A  LOAD=OFF  STATE=DISCONNECTED  FAULT=UNDER-VOLTAGE  TRIPS=1
[30271 ms] V=80.1 V  I=6.22 A  LOAD=OFF  STATE=DISCONNECTED  FAULT=UNDER-VOLTAGE  TRIPS=1
[31287 ms] V=80.1 V  I=6.22 A  LOAD=OFF  STATE=DISCONNECTED  FAULT=UNDER-VOLTAGE  TRIPS=1
```

High Current Condition:

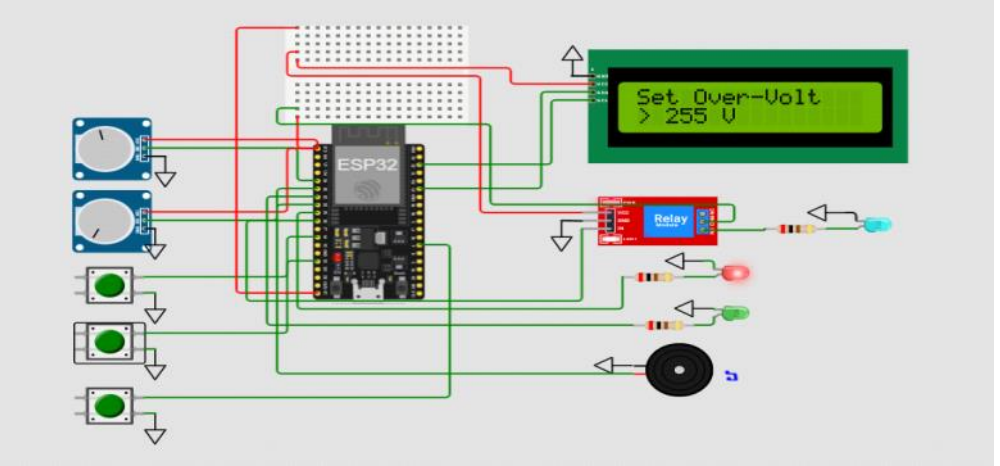


Result In Serial Monitor:

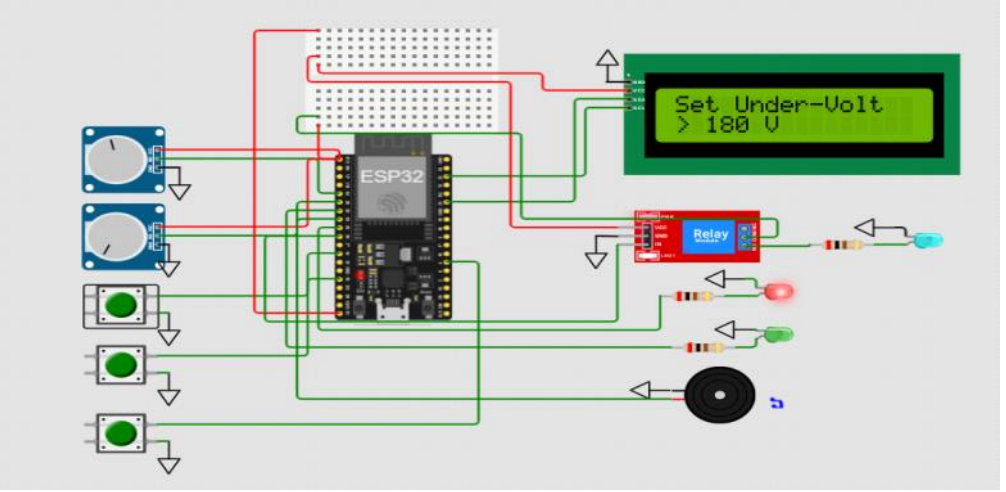
```
[44476 ms] V=201.2 V  I=4.89 A  LOAD=OFF  STATE=DISCONNECTED  FAULT=OVER-CURRENT  TRIPS=2
[45492 ms] V=263.4 V  I=4.89 A  LOAD=OFF  STATE=DISCONNECTED  FAULT=OVER-CURRENT  TRIPS=2
[46508 ms] V=235.5 V  I=4.89 A  LOAD=OFF  STATE=DISCONNECTED  FAULT=OVER-CURRENT  TRIPS=2
[47524 ms] V=235.5 V  I=4.89 A  LOAD=OFF  STATE=DISCONNECTED  FAULT=OVER-CURRENT  TRIPS=2
[48540 ms] V=235.5 V  I=4.89 A  LOAD=OFF  STATE=DISCONNECTED  FAULT=OVER-CURRENT  TRIPS=2
[49556 ms] V=235.5 V  I=4.89 A  LOAD=OFF  STATE=DISCONNECTED  FAULT=OVER-CURRENT  TRIPS=2
[50572 ms] V=235.5 V  I=4.89 A  LOAD=OFF  STATE=DISCONNECTED  FAULT=OVER-CURRENT  TRIPS=2
```

Settings Changes:

1) High Voltage settings



2) Low Voltage Settings



3) Over Current Settings

